Scene-to-Audio: Distant Scene Sonification for Blind and Low Vision People

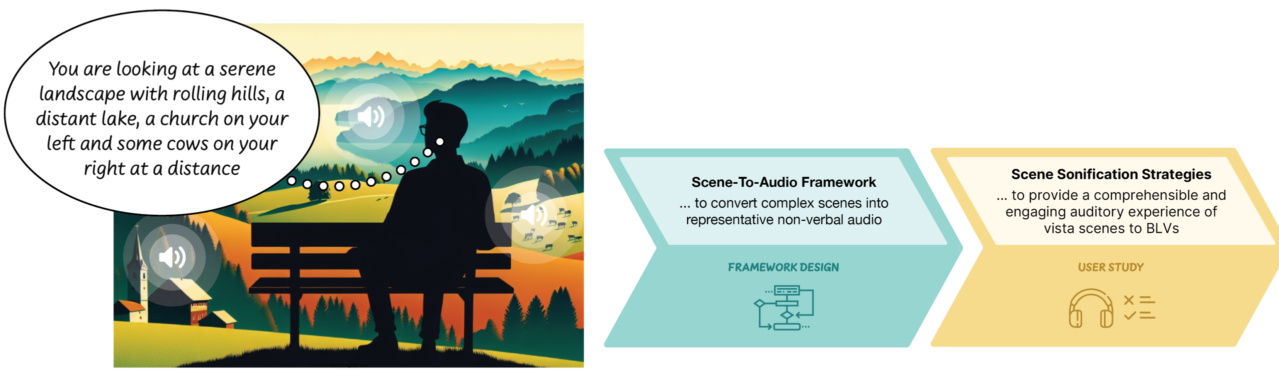
Chitralekha Gupta
Augmented Human Lab, School of Computing
National University of Singapore
Singapore, Singapore
chitralekha@nus.edu.sg

Ashwin Ram
Saarland Informatics Campus
Saarland University
Saarbrücken, Saarland, Germany
ashwinram10@gmail.com

Shreyas Sridhar
School of Computing
National University of Singapore
Singapore, Singapore, Singapore
shreyas@comp.nus.edu.sg

Christophe Jouffrais
IPAL
CNRS
Singapore, Singapore
IRIT
Univ of Toulouse
Toulouse, France
christophe.jouffrais@cnrs.fr

Suranga Nanayakkara
Augmented Human Lab
School of Computing, National University of Singapore
Singapore, Singapore
suranga@ahlab.org



(a) (b)

Figure 1: (a) A person with visual impairment able to enjoy a distant environmental scene through comprehensible and enjoyable sounds, i.e. a combination of verbal description and non-verbal representative sounds. (b) In this paper, we developed a Scene-to-Audio framework and utilized it to create an auditory experience of Vista scenes for BLV.

Abstract

Awareness of distant environmental scenes, or Vista scenes, is necessary for comprehending one's surroundings and enjoying their beauty in leisurely settings. Current tools for Blind and Low Vision people (BLV) typically use spoken descriptions but lack support for an enjoyable experience of visually pleasing scenes. We propose a Scene-to-Audio framework that generates comprehensible and enjoyable non-verbal sounds representing distant scenes using generative models informed by psycho-acoustics, audio scene analysis, and Foley sound synthesis. We conducted a user study with eleven BLV participants evaluating strategies to utilize our scene-to-audio framework. We found that these sounds, in combination with speech, provide a significantly more memorable and immersive experience of the scenes, compared to only spoken descriptions. Our work is a step towards bridging the gap between a visual and an auditory experience of a scene, addressing the aesthetic needs of BLVs.

CCS Concepts

• **Human-centered computing** → **Empirical studies in accessibility**; **Accessibility technologies**.

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Keywords

Blind and Low Vision, People with Visual Impairments, Vista Scenes, Spatial Cognition, Scene Experience, Cognitive and Aesthetic Needs, Sonification, Generative Models for Audio

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1 Introduction

Distant environmental scenes, that are beyond audible range but within visual range, belong to the "Vista" space [39]. Enjoying views on a rooftop or after hiking to the top of a hill are examples of such vista scenes. This experience is crucial and contributes to one's cognitive and aesthetic needs, situated in the upper half of Maslow's Hierarchy of needs [20, 35, 38]. Arguably, experiencing vista scenes may go beyond the purpose of comprehending the scene, especially in the context of leisure, where one seeks the pleasure of experiencing the views and appreciating the beauty of the scenery. However, this type of experience is not easily accessible to Blind and Low Vision People (BLV)[1].

Prior work has established that building an understanding of distant environmental scenery such as visual landmarks as well as enjoying scenery is important for BLV [4, 17, 20]. This is found to be especially true amongst people who developed their visual impairment condition later in life, as they often have enjoyable memories that these views once provided, which they now miss [4, 20]. Increasing attention is being given to the cognitive and aesthetic needs of BLVs, such as providing accessible media content [33, 37, 45], sports content [24], visual arts [31], and mixed reality experiences [2, 9, 36, 54]. Although some solutions use the state-of-the-art generative models to offer textual or spoken descriptions of scenes, these spoken descriptions alone can impose a high cognitive load on BLVs, and also interfere with other auditory inputs. This may lead to a less satisfying experience compared to the original visual experience [13, 25]. Non-verbal sounds, on the other hand, have natural auditory associations that make them intuitive and fast to comprehend [1, 11, 15].

In this paper, our main aim is to design a framework that can automatically generate non-verbal sounds corresponding to vista scenes, that provides scene understanding as well as an enjoyable experience to BLV. For this, we propose a framework, "*Scene-to-Audio*", which has a two-step process. In the first step, it generates non-verbal representative audio for the different objects in the scene (*Salient Objects Identification*). In the second step, it combines those individual audio components to create a representative audio for the scene, informed by psycho-acoustics, auditory scene analysis, and Foley sound synthesis [8, 10, 40] (*Audio Scene Composition*).

[1]We understand it is important to use neutral and accurate language to describe people with disabilities, as it promotes respect and inclusivity. Following the recommendations from the SIGACCESS community [21], we use the term *"Blind and Low Vision (BLV)"*, when referring to individuals with vision loss. The use of this abbreviation is meant for ease of reference and consistent with prior work [17, 26]. We also acknowledge that other terms might also be appropriate [46], such as *"People with Visual Impairments (PVI)"*.

Through a listening test with 21 sighted participants, we evaluate the sounds generated from *Scene-to-Audio* framework in terms of their closeness of representing a vista image. We find that the sounds from our framework outperform those generated with existing baseline techniques. We then investigate different strategies to utilize our *Scene-to-Audio* framework to provide an engaging experience of a distant scene to BLVs. We conduct a user study with 11 BLV participants to evaluate these strategies. We find that overlaying verbal description with representative non-verbal sounds is significantly more effective than only the verbal description of the scene in providing a memorable and immersive experience, while being highly comprehensible and low in cognitive load. Specifically, our contributions in this paper are two-fold:

- We present a novel framework, *Scene-to-Audio*, to convert scene images to comprehensible sounds,[2], and
- Through a user study with BLV participants, we evaluate strategies to utilize the sounds generated by our framework and present insights into designing more inclusive and engaging vista scene experiences for BLVs.

2 Background & Related Work

2.1 Role of non-verbal audio for scene experience

In audio stories such as those on radio, podcasts, and audiobooks, where visual imagery is absent, empirical studies have shown that sound effects are beneficial. Specifically, sound effects serve three main purposes. First, they reinforce the actions that occur in the scene, such as a car whooshing by. Second, they simulate the atmosphere of the narrative to immerse or transport the listener into the scene, like the sound of traffic on the street [18, 48]. Lastly, sound effects are used to evoke emotions, for example, the sound of rain to evoke a feeling of melancholy [48]. Consequently, non-verbal sound effects, particularly when visual imagery is absent, are essential for scene awareness. They not only help in understanding the scene but also enhance immersion and the overall listening experience, leading to an augmented sense of hearing. This inspired us to develop techniques to automatically generate non-verbal sound effects for vista scenes, aimed at enhancing scene awareness and experience for BLVs.

As Crisell [12] and Fryer [14] stated, sound effects – when used effectively – can stimulate our imagination. But, to convey clear meaning, verbal description is necessary along with the sound effects. In the context of enhanced audio descriptors (EAD) for movies designed for BLVs, Lopez et al. [34] and Jiang et al. [26] demonstrated that audio effects complement verbal descriptions, thereby reducing cognitive load, and Hattich et al. [23] showed that such audio effects improve immersion. Additionally, combining audio with speech during storytelling has shown higher emotion induction and transportation into the scene compared to speech alone amongst BLVs [48]. Therefore, we design strategies to combine verbal descriptions with AI-generated audio effects for real-world vista scene awareness for BLVs, and validate their effectiveness in providing immersive and experiential effects.

[2]our source code will be made available upon acceptance

2.2 Non-verbal sounds in assistive devices for BLV

Recently, there has been a growing focus on developing assistive devices and applications for BLVs that go beyond basic needs like obstacle detection and navigation. Increasing attention is being given to the cognitive and aesthetic needs of BLVs, such as providing accessible media content [33, 37, 45], sports content [24], visual arts [31], and mixed reality experiences [2, 36, 54] through both verbal and non-verbal sounds. Prior research has also emphasized the importance of outdoor nature scene awareness and experience in fulfilling these cognitive and aesthetic needs in BLVs [3, 4, 20]. We believe that experiencing the aesthetics of a vista scene involves not only communicating information about the objects in the scene but also providing an engaging experience.

Non-verbal audio, such as auditory icons, is often used in assistive applications for BLVs for object awareness [11, 50]. However, synthetically designing auditory icons requires parametric sound modeling, which involves designing sounds for individual objects. This process lacks scalability due to the manual crafting of model parameters, which demands expert knowledge. This becomes more challenging when a scene needs to be sonified, as it may consist of multiple objects simultaneously present in the field of view. In such cases, generative models present a promising solution. Gupta et al. [20] found that the sounds for specific objects generated from current AI-based audio generative models are comparable in quality and intuitiveness to expert-designed auditory icons for those objects. However, there is a significant degradation in the audio quality of AI-generated sounds for scenes with multiple objects or events compared to those for scenes with one object or event. This issue arises because these AI models are predominantly trained on datasets consisting of objects and event labels, rather than on scene soundscapes that include multiple objects or events occurring simultaneously. In this work, we address this gap by designing a framework built upon existing AI models that can automatically compose an auditory scene for a vista scene image, making it comprehensible and enjoyable.

2.3 AI techniques for Audio Generation

Designing generative models that convert videos [53], images [47], and text [32] to representative non-verbal audio are active areas of research. In video-to-audio models, the task is to generate audio effects for a silent video where an action occurs. In this work, we consider distant environmental scenes where actions occurring in the far field are not expected to be prominent. Hence we consider a static image of the scene to be a closer approximation of a distant environmental scene. The state-of-the-art image-to-audio (non-verbal) model, Im2Wav [47], consists of a transformer-based audio generative model that is conditioned on image representations obtained from a pre-trained CLIP model [43]. Text-to-audio (non-verbal) models such as AudioGen [29] and AudioLDM [32] are able to generate high-quality audio from textual inputs. For example, AudioGen is a state-of-the-art autoregressive, transformer-based audio generation model, that generates audio from text by using textual features as conditioning signal. Some recent works have demonstrated the use of such AI models for different applications, such as generating sounds for individual objects in an augmented reality experience [49]. These general-purpose image-to-audio and text-to-audio models are primarily trained on large datasets of isolated objects and event labels. As a result, applying these models directly to complex scenes with multiple objects or events may yield sub-optimal results.

Existing mobile applications to help BLVs to be aware of their surroundings are typically image-to-text-to-speech frameworks. For example, SeeingAI[3] and BeMyAI[4] in the BeMyEyes app, utilize state-of-the-art image-to-text models like OpenAI's ChatGPT-4[5] to provide textual descriptions of images. These descriptions are then converted to speech using text-to-speech accessibility features such as VoiceOver[6] on iPhones and TalkBack[7] on Android devices. Gonzalez et al. [17] investigated the use-cases and motivations of BLVs when using "scene description" or image-to-text apps through a two-week diary entry study with BLVs. They found that one of the top three goals of using scene description app amongst BLVs is for "building understanding of the scenery". While speech provides direct information, it often results in a high cognitive load and can interfere with other voice inputs [13, 25]. Thus, in this work, we explore alternate methods of sonification for the task of understanding and experiencing scenes.

3 Scene-to-Audio framework for Automatic Vista Scene Sonification

In this paper, our goal is to convey the key elements and overall atmosphere, or *"gist"*, of a vista scene through a representative non-verbal audio [22, 42]. A straightforward approach would be to use a state-of-the-art generative model such as Im2Wav [47] to convert an image to an audio clip. Such models are typically trained for one object or event in focus. Vista scenes, on the other hand, often have several objects in the scene, which can be difficult to sonify using existing image-to-audio models. Moreover, training a model on scene data is difficult due to a lack of existing datasets that consist of images of complex vista scenes and corresponding audio that conveys its gist.

To address these challenges, in our proposed Scene-to-Audio framework, we identify salient objects from the scene image and optimize how their corresponding sounds are combined. These enhancements are designed to create a more enjoyable and immersive vista experience, guided by psycho-acoustics, auditory scene analysis, and Foley sound synthesis. Our Scene-to-Audio framework include two components. The two components are - (1) Salient Objects Identification, and (2) Audio Scene Composition. The function of the first component is to analyze a distant scene image using an image-to-text model (we use GPT-4) to identify key sound-making or sonic objects within the scene. These objects are described by action phrases (noun+verb), which serve as text prompts for a text-to-audio model in the second component which then layers the generated sounds to create a balanced and immersive auditory experience. We employ prompt chaining technique[8] across the two AI models (image-to-text and text-to-audio). The image-to-text model

[3] https://www.seeingai.com/
[4] https://www.bemyeyes.com/blog/introducing-be-my-eyes-virtual-volunteer
[5] https://openai.com/index/gpt-4/
[6] https://support.apple.com/en-sg/guide/voiceover/welcome/mac
[7] https://support.google.com/accessibility/android/answer/6283677?hl=en
[8] https://www.promptingguide.ai/techniques/prompt_chaining

is prompted with a sub-task and then its response is used as an input prompt for the text-to-audio model. Figure 2 provides the details of the two components of the proposed framework.

3.1 Salient Objects Identification

To extract sonic objects from the scene image, we use the following prompt with GPT-4: *"Describe this image with respect to the sound-making objects in the scene."*. Once the sonic objects are identified, generating appropriate sounds for these objects requires a carefully designed set of steps as follows:

3.1.1 Generate Action Phrases. A fundamental difference between visual perception and auditory perception is how we perceive objects. We see visual objects, but we hear *sound events* [15, 41]. While sounds may originate from an object (such as a person or a vehicle), they are triggered when an event on or by the object occurs. For example, leaves (the object) produce sound when they are moved by the wind (the event). Thus, we associate each identified sonic object with an action phrase — a noun paired with a verb. The noun represents the object, and the verb represents the action that produces the sound. We prompt GPT-4 with *"Only provide a noun+verb like phrase such that the noun is the sound making object and the verb is the possible action on or by the object that could create a sound"*.

3.1.2 Identify the Sound Event Type. The human auditory system processes different types of sound events—continuous and discrete—differently [40]. Continuous sounds, such as wind or ambient noise, typically form the background of an auditory scene and are perceived as less salient. In contrast, discrete sounds, such as a bell ring or a door slam, are more likely to capture attention due to their distinct temporal characteristics. To provide the subsequent Audio Scene Composition component with this additional control, i.e. the *event-type* label of the sound, we prompt GPT-4 with: *"At the end of each noun + verb phrase, attach a label "discrete" or "continuous", depending on whether the sound produced by the object is generally discrete or continuous"*. The complete text prompt to the image-to-text model is provided in Figure 2a.

3.2 Audio Scene Composition

In distant scenes, the exact spatial positioning of objects may be less critical for understanding the scene's overall auditory representation [8, 16]. Therefore, in the Audio Scene Composition component of our framework, we assume that all objects in the distant vista scene are equidistant from the user's point of view. This simplification allows us to focus on conveying the essence of the scene through sound rather than on the precise spatial relationships between objects. Figure 2b presents details of this component, and the key decisions are explained below.

3.2.1 Differentiating Discrete and Continuous Sounds. We treat discrete and continuous sounds differently as this distinction is crucial for achieving a realistic and immersive auditory experience, which is grounded in psychoacoustics and auditory scene analysis [8, 40]. Discrete sounds, such as footsteps or a car horn, are characterized by isolated audio events with clear onsets and offsets. These sounds naturally draw attention and are perceived as being in the foreground, regardless of their actual distance from the listener. In contrast, continuous sounds, like the hum of a distant highway or the rustling of leaves, create a persistent background layer. These are generally not the main focal point of the scene and are perceived as ambient elements. Thus we inform the audio scene composition component with the event-type label (i.e. discrete or continuous) generated in the previous step.

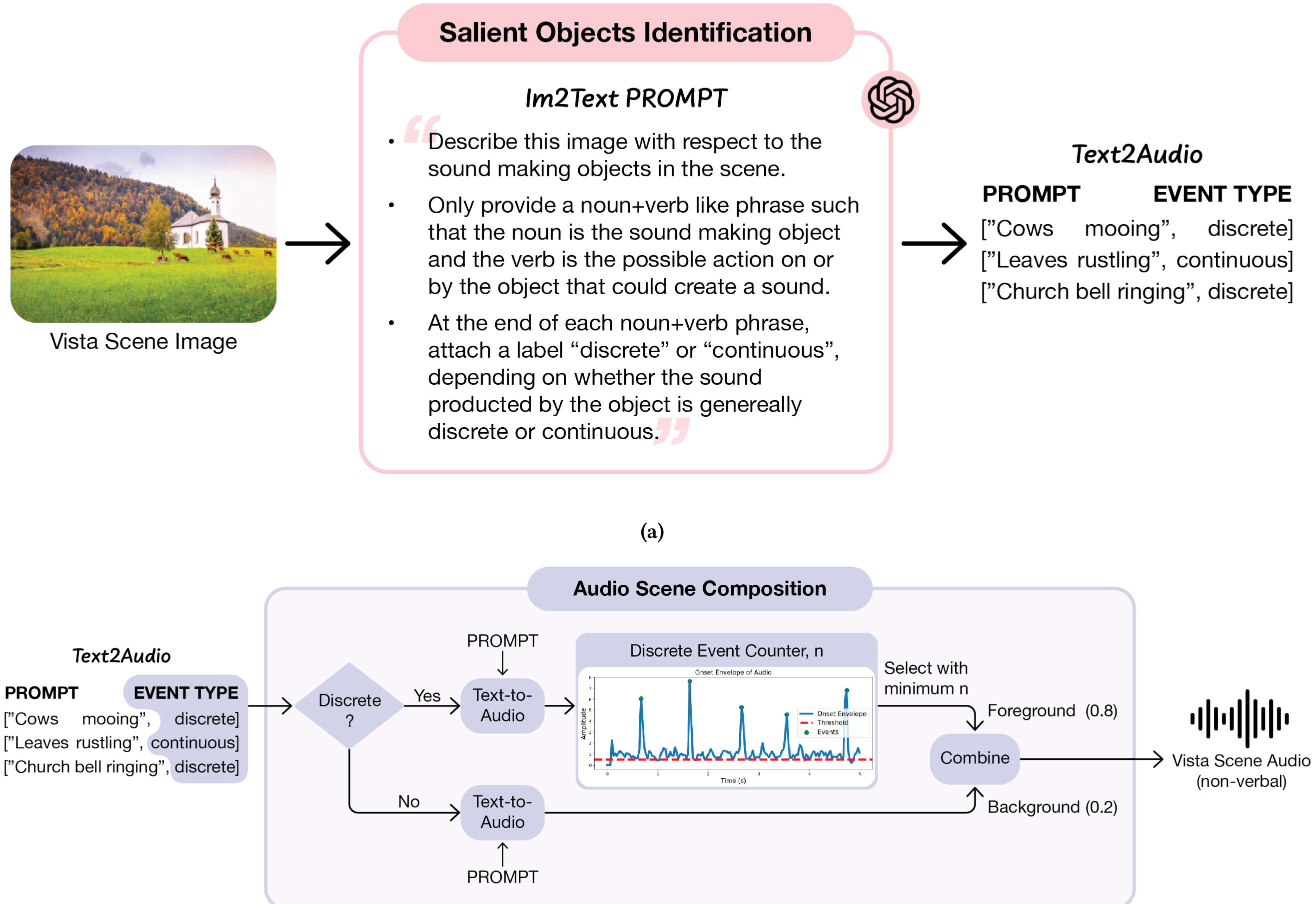
3.2.2 Minimizing Discrete Sound Events. Psychoacoustic research indicates that repetitive, high-intensity discrete sounds can lead to discomfort and auditory fatigue [30]. Continuous exposure to such sounds can result in auditory masking [40], where these prominent sounds overshadow other auditory cues, reducing the clarity and immersiveness of the experience. To prevent these issues, we incorporate a mechanism to minimize the number of events in the generated discrete sounds.

In current text-to-audio models, there is no straightforward way to control the number of discrete events in the generated sound as these models are trained on audio with captions that do not elicit this granularity in semantics [19, 27, 28, 52]. For example, a typical text label would be "a stick hitting on a wooden table", but it would not mention the frequency of hits or the number of hits. Thus, attempts to use different text prompts to manage the number of events yield inconsistent results with these models [28]. Therefore, we adopted a signal processing-based method to achieve the desired control. Specifically, when a text prompt is tagged as "discrete", we generate ten audio clips, each five seconds long, using the text-to-audio model (AudioGen). We then analyze each audio clip using peak detection in the amplitude envelope. This method allows us to count the number of discrete events within each clip and select the one with the least number of events n ($n \geq 1$). This increases the likelihood that the chosen sound is not overly repetitive or jarring, thereby providing a more comfortable listening experience.

3.2.3 Foreground and Background Layering. To create a balanced auditory scene, we perform a weighted average of all the sounds, assigning different weights to discrete and continuous sounds. Discrete sounds are placed in the foreground using a higher weight (0.8), while continuous sounds are placed in the background with a lower weight (0.2). This approach is informed by Foley sound synthesis techniques [5, 10], which create sound effects for media such as movies to enhance the realism of visual scenes. In Foley sound synthesis, the foreground sounds are designed to capture the listener's attention and provides key information for immediate understanding of the scene, while background sounds add context, depth, and immersion without competing for focus.

4 Utilizing Scene-to-Audio Framework for Enhancing Scene Experience of BLVs

Our evaluation showed that non-verbal audio generated from Scene-to-Audio framework can create comprehensible and pleasant experiences. However, the non-verbal sounds alone can sometimes lead to ambiguity, as the sounds may have multiple interpretations. On the other hand, verbal descriptions, while being clear, often lack engagement. Here, we explore strategies to combine non-verbal audio with verbal descriptions for a richer, more immersive experience.



(b)

Figure 2: Detailed Scene-to-Audio Framework consisting of the two components, (a) Salient Objects Identification, and (b) Audio Scene Composition.

4.1 Scene Sonification Strategies for BLVs

First, we investigated the impact of auditory feedback variants that combine non-verbal and verbal sounds, aiming to enhance scene comprehension and user experience. Second, we explore how providing more detailed information about the scene, including the relative positioning of various objects, could influence the user's comprehension of the scene, cognitive load, and overall immersion. We design four strategies of audio feedback for a given vista scene, which we call *Audio types*:

- **Speech-only**: Speech or spoken description is considered as the baseline approach that represents the auditory feedback mechanism of existing apps for scene understanding (eg. SeeingAI and BeMyAI) used by BLVs. It consists of a short spoken description of the scene.
- **Audio-only**: This is the non-verbal representative audio of a scene generated from our Scene-to-Audio framework.
- **Overlay**: The verbal description (speech-only) is overlaid on the non-verbal sound (audio-only) of the scene, with the verbal description in the foreground.
- **Overlay-Concat**: Sonifying different sonic objects in the scene and overlaying each one with the corresponding speech description. They were then played sequentially. This included more details about the scene along with the relative positions of the different sonic objects in the scene with respect to the user. For example, for the vista scene image in Figure 2a, it would generate the following overlay of text-to-speech and scene-to-audio: *"A group of cows is directly ahead in a lush, green meadow."*; followed by *"To your right, trees with leaves in autumn hues are on a slope."*; followed by *"A small white church with a steeple sits in the distance, slightly to the left."*.

Readers are encouraged to listen to these four Audio Types for all the scenes on our webpage[9].

[9]https://scenetoaudio.github.io/scenetoaudio/#/section-5

4.2 Study Design

We used a within-subject design to compare the four audio types for different vista scenes. The order of audio types was counterbalanced to avoid sequence bias.

4.2.1 Participants. The study involved 11 blind and low vision participants (BLVs), including 9 males and 2 females, ranging in age from 27 to 73 years old (mean age 49.8 years, std. dev 17.14). The participants had a variety of visual conditions, such as Retinitis Pigmentosa, Glaucoma, and Macular Degeneration. Most participants developed visual impairments after the age of 14, with the exception of one participant (P4) who has been visually impaired since birth. Visual acuity varied amongst the participants, with most reporting significant vision loss (e.g., less than 2% to 10% visual acuity). Detailed demographics of the participants are in Appendix. The participants were recruited through local organizations for the visually impaired. We obtained ethics approval from the Institute Review Board (IRB) before conducting the user studies.

4.2.2 Scene images. For this evaluation, we selected a diverse set of 8 vista scene images, split evenly between natural and urban environments. The natural scenes included a *seabeach*, *mountains*, *countryside*, and a *reservoir*, while the urban scenes comprised an outside *train station*, *park*, *food court*, and a *street*. The scenes were further categorized based on the density of sonic objects: the seabeach, mountains, train station, and reservoir featured around 2 to 3 sonic objects, representing sparser auditory environments. In contrast, the countryside, park, food court, and street contained 4 to 5 sonic objects, offering more complex auditory challenges. This variety of scenes and object densities allowed us to comprehensively assess the effectiveness and adaptability of our Scene-to-Audio framework in handling different auditory environments.

4.2.3 Listening Test Design. We generated the four audio types - Audio-only, Speech-only, Overlay, and Overlay Concat (section 4.1), for all the eight scenes.To ensure a fair comparison between audio types, we used different scenes for each audio type and counterbalanced the scene-audio type combination across participants. This approach helped minimize the impact of inherent information differences in scenes between conditions (eg. Overlay having more information than Audio-only).

The listening test was conducted in a controlled lab environment. Participants were asked to imagine themselves on a hilltop or rooftop, using a system that converts distant scenes into audio. Most participants were familiar with AI-based image-to-text apps like BeMyEyes, Seeing AI, and Google Lookout (see Appendix), which aided their understanding of the additional sound effects and engage with the experiment.

As indicated in the pilot study, participants preferred using their usual earphones. We provided a selection of headphone options, allowing participants to choose their preferred headphones during the experiment. The Google Text-to-Speech engine used in the pilot study was perceived as robotic. For the main user study, we switched to Eleven Labs' text-to-speech engine[10], which offers more natural, human-like speech.

[10]https://elevenlabs.io/text-to-speech

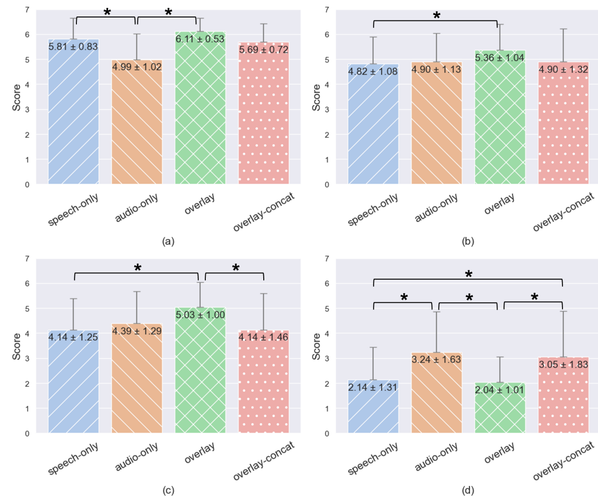


Figure 3: Overall ratings averaged across questions for each criteria - (a) Comprehension (higher is better), (b) Engagement (higher is better), (c) Immersion (higher is better), and (d) Cognitive Load (lower is better). Pairwise statistical significance using Wilcoxon signed rank test is shown with a * for significantly different pairs ($p < 0.05$).

4.3 Measures

We asked the participants to complete a questionnaire (administered verbally) after listening to each audio sample. This questionnaire consisted of evaluating four key criteria: *Comprehension*, *Engagement*, *Immersion*, and *Cognitive Load*. These criteria were chosen based on the aspects of vista scene experience indicated as important in previous works [8, 17, 20]. We carefully adopted questions from existing questionnaires assessing these criteria in literature.

At the end of listening to the eight audio clips, we asked participants to provide a rank-order of their preference of the 4 audio types they heard. We also asked a few open-ended questions about their audio-type preference etc. Please refer to the Appendix for the full details of questionnaire.

4.4 Results

The four criteria—Comprehension, Engagement, Immersion, and Cognitive Load —were evaluated across the four different audio types: Speech-only, Audio-only, Overlay, and Overlay Concat. We first conducted Shapiro-Wilk normality test and found that the normality assumption is not met for all variables. Thus, the results were analyzed using the non-parametric Friedman test for main effect, and the non-parametric Wilcoxon signed-rank test was used for post-hoc analysis. The overall scores, i.e. average scores across questions for each criteria is visualized in Figure 3, with statistically significant (Wilcoxon signed rank test, $p < 0.05$) comparisons are marked with a $*$. The detailed average values for each question (mean ± standard deviation) is provided in Appendix.

Across the overall comprehension scores for the four audio types, Friedman test ($\chi^2(3) = 8.2, p = 0.04$) revealed a statistically significant main effect. The post-hoc Wilcoxon signed-rank test showed

Speech-only (5.81±0.83) and Overlay (6.11±0.53) significantly outperformed Audio-only (4.99±1.02) ($p = 0.0093$ and $p = 0.0009$ respectively).

The Overlay audio type showed significantly better results in terms of enjoyability, scoring higher than both Audio-only and Overlay Concat audio types ($p = 0.002$ and $p = 0.03$ respectively). The curiosity related question saw the Audio-only condition scoring highest, significantly outperforming the Speech-only condition ($p = 0.04$). The Imagination related question had the Overlay condition with the highest score, though no significant differences were noted. Finally, for the Memorability related question, the Overlay condition again had the highest score, significantly higher than Speech-only ($p = 0.04$). Detailed results in Appendix.

The Overlay condition consistently scored the highest across all factors of Immersion criteria - Presence, Spatial Presence, Involvement, and Experienced Realism (please refer to Appendix). Notably, Experienced Realism in the Overlay audio type was significantly higher than Speech-only ($p = 0.0049$), Audio-only ($p = 0.039$), and Overlay Concat ($p = 0.008$). Overlay also outperformed Speech-only for Spatial Presence with statistical significance ($p = 0.04$).

Regarding Cognitive Load, lower scores indicate better performance. Friedman test indicated statistically significant main effect in Cognitive Load across the audio types ($\chi^2(3) = 9.8, p = 0.02$). Post-hoc analysis revealed that Audio-only (3.24±1.62) and Overlay-Concat (3.05±1.83) had a significantly higher cognitive load compared to Speech-only (2.14±1.31) ($p = 0.02$ and $p = 0.04$ respectively) and Overlay (2.04±1.01) ($p = 0.01$ and $p = 0.008$ respectively).

At the end of listening to the four audio types, each participant was asked to rank order the four audio types according to their preference. Overlay is the most preferred audio type, with a positive inter-rater agreement (Kendall coefficient $\tau = 0.3, p = 0.0$). The average preference scores are shown in Appendix.

The qualitative feedback, analyzed using thematic analysis per Braun and Clarke's guidelines [7], revealed themes aligning with the quantitative results. Overlay audio (speech combined with non-verbal sounds) was generally preferred for enhancing comprehension, immersion, and engagement, making the experience more memorable. For instance, P1 described overlay sounds as *"a lot more pleasant to listen and it was easier to imagine the scene"*, while P6 noted, *"The sound effects create movement in the scene, and stirs emotion, and imagination"*. In contrast, speech-only audio, though easy to comprehend, was perceived as less immersive, with P9 remarking, *"As there are no sound effects, I don't feel I am present there"*. Sequential presentation of elements (Overlay Concat) provided clear directionality but was considered less immersive and more cognitively taxing, highlighting the need for smoother transitions. Participants emphasized the complementary roles of speech and non-verbal audio, with P7 stating, *"The distinctive sounds really help in imagining the scene and bring you into the scene, while the narration is very helpful in giving details"*, and P1 summarizing, *"[Non-verbal] sound sets the stage, and speech guides you along the scene"*.

5 Discussion and Future Work

Our study highlights the value of combining verbal descriptions with non-verbal sound effects from our Scene-to-Audio framework to enhance the vista scene experience for BLVs, significantly improving memorability, presence, and enjoyment compared to speech-only descriptions. This aligns with prior research showing that non-verbal sounds aid in mental imagery, evoke emotions, and enhance immersion [6, 44, 48]. However, sound effects alone are insufficient for clear communication, as speech is essential for guidance and disambiguation, particularly when sounds carry varied cultural or contextual meanings, such as traffic light beeps. Additionally, our framework's structured approach, rooted in psycho-acoustics and auditory scene analysis [8, 40], enables fine-tuned sonic object modeling and layering, addressing limitations of baseline generative models in complex scenes. Guided by principles of mental model alignment and co-creation [51], our design ensures inclusivity, accessibility, and engaging leisure experiences for BLVs by tailoring sonification strategies to their needs.

Our study provides valuable insights into the effectiveness of the Scene-to-Audio framework for conveying vista scenes to individuals with visual impairments, but several limitations and avenues for future work remain. The small number of BLV participants limits the generalizability of our findings, though their diverse visual conditions and consistent feedback offer meaningful design implications for assistive technologies. Testing in controlled environments may not fully capture real-world dynamics, highlighting the need to develop a system for real-world evaluation, address practical challenges like latency, and explore edge computing for faster processing. While the framework demonstrated scalability with a carefully selected range of scenes, broader testing across diverse and complex scenes is needed to assess generalizability, constrained currently by the lack of large, annotated datasets. Finally, ensuring seamless control over discrete events in generated audio remains a challenge, and future work should focus on integrating more precise control mechanisms into generative models to improve sound quality and adaptability.

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